

Course Title:	Categorical Data Analysis
Course Code:	STAT-416
Semester:	VII
Credit Hours:	3 Credit Hours
Pre-requisites:	N / A

Learning Outcomes

By the end of this course, students will be able to:

1. Know the nature of categorical variables and their measurement scales.
2. Ascertain the knowledge of sampling distributions for categorical variables and learn the methods of testing hypothesis with applications.
3. Learn various tools and techniques to analyze categorical data and their applications to the real world examples.

Course Outlines

Unit 1

1.1 Introduction

Historical background of categorical response data. Types of categorical variables, measurement scales distinction and sampling distributions. Testing goodness of fit and independence. Large sample confidence intervals and the idea of p-value.

1.2 Different Test statistics

Chi-square Test for Categorical Data, its Assumptions and Applications, Phi Co-efficient, Contingency Co-efficient (C), Cramer's-V, Adjusted Chi-square (Yates' Correction), Fisher's Exact Test (An Exact Test for (2x2) Contingency Table), Kendall's Tau b Statistic. Cochran-Mantel-Haenszel Test, Matched Samples and McNemar Test. Meta-Analysis for (2x2xk) Tables.

Unit 2

2.1 Testing of Statistical Significance

Testing of Statistical Significance of Relative Risk and Odds Ratio with its Confidence Limits, Sensitivity, Specificity and Kappa Statistic.

2.2 Model building

Models for Binary Response Variables, Log Linear Models and Fitting of Log-linear and Logit Models, Binomial and Normal Probability Plots.

Text Book

1. Agresti, A. (2012). *Categorical data analysis* (3rd ed.). John Wiley & Sons.

Suggested Readings

1. Lloyd, C. J (1999). *Statistical analysis of categorical data* (1st ed.). John Wiley & Sons.
2. Powers D. A., & Xie, Y. (2008). *Statistical Methods for Categorical Data Analysis* (2nd ed.). Emerald Group Publishing.

BS (4 Years) for Affiliated Colleges



Code	Subject Title	Cr. Hrs	Semester
STAT-401	Statistical Inference-I (Theory)	3	VII
Year	Discipline		
4	Statistics		

Code	Subject Title	Cr. Hrs	Semester
STAT-402	Statistical Inference-I (Practical)	1	VII
Year	Discipline		
4	Statistics		

Course Outline

Point estimation, problem of estimation. Properties of a good estimator: Unbiasedness, Consistency, Efficiency and Sufficiency. Mean-squared error. Consistency and Best asymptotically normal estimator. Minimal sufficient statistics. Joint sufficiency. Exponential family. Sufficiency and Completeness. Cramer-Rao inequality. Minimum Variance Bound estimators. Rao-Blackwell and Lehmann-Sheffe theorems. Uniformly Minimum Variance Unbiased estimators. Joint completeness. Location Invariant and scale-Invariant estimators. Pitman estimators for location and scale.

Bayes estimators. Prior and Posterior distributions. Posterior Bayes estimators. Loss function and Risk function. Bayes estimator, Minimax Methods of estimation.

Recommended Books:

- Hogg, R.V. and Craig, A.T. "Introduction to Mathematical Statistics", Prentice-Hall International, Inc. Engle Wod Cliff, N.J., Sixth Edition, 2004.
- Hogg, R.V. and Tanis E.A., "Probability and Statistical Inference" Macmillan Publishing Company, New York, Seventh Edition, 2009.
- Mood, A.M. Graybill, F.A. and Boes, D.C., "Introduction to the Theory of Statistics", McGraw-Hill Book Company, New York, Third Edition, 1974.
- Levy, P.S. and Lemeshow, S., "Sampling of Populations: Methods and Applications", John Wiley, New York, Third Edition, 1999.
- Lehman, E.L. "Theory of Point Estimation", John Wiley, New York, 1983.
- Rao, C.R., "Linear Statistical Inference and its Applications", John Wiley, New York, 1973.
- Hoel, P.G. "Introductions to Mathematical Statistics" Fifth Edition, John Wiley, 1984.

BS (4 Years) for Affiliated Colleges



Code	Subject Title	Cr. Hrs	Semester
STAT- 403	Basic Econometrics (Theory)	3	VII
Year	Discipline		
4	Statistics		

Code	Subject Title	Cr. Hrs	Semester
STAT-404	Basic Econometrics (Practical)	1	VII
Year	Discipline		
4	Statistics		

Course Outline

1. Econometrics

Its nature, methodology and functions.

2. Simple Linear regression

Ordinary least squares method; assumptions and estimation. Maximum likelihood method; assumptions and estimation, Properties of OLS and ML estimators. Partition of total sum of squares. Sampling distribution of sum of squares, Testing of hypotheses confidence intervals for the parameters and Linear combinations of parameter. Comparison of simple linear regressions, Chow test.

3. General Linear regression

Ordinary least squares method; assumptions and estimation. Maximum likelihood method; assumptions and estimation properties of OLS and ML estimators. Partition of total sum of squares. Sampling distribution of sum of squares, Testing of hypotheses for the single, all some any parameters, Linear combinations of parameters. Comparison of general linear regressions, Chow test. Gauss Markov's theorem.

4. Other topics

Stepwise regression, Ridge regression, GLR partitioned form: Estimator & testing of hypothesis, Use of extraneous information in linear regression. Restricted least squares estimator.

BS (4 Years) for Affiliated Colleges



Code	Subject Title	Cr. Hrs	Semester
STAT-411	Time Series Analysis-I	3	VII
Year	Discipline		
4	Statistics		

Course Outline

1. Introduction to time series, time series analysis, Objectives of time series analysis, Components of time series, time series plots, time series and stochastic processes, special features of time series data, means, variance, auto-covariance, auto-correlation and partial auto-correlation for sample time series data.
2. Simple Descriptive Techniques: Stationary time series, transformations, Analyzing the secular trend, Filtering, Differencing, Analyzing Seasonal Variations, Analyzing Cyclical Variations, Analyzing Irregular Variations, Auto-correlation (correlogram) and other tests of randomness.
3. Probability Models for Time Series: Stochastic processes and stationary processes, useful stochastic processes, purely random process, random walk, moving average process, Stationarity and Invertibility of moving average models, auto-regressive process, Stationarity and invertibility of auto-regressive models, Duality between moving average and auto-regressive models, Principle of parsimony, Recursion rule for ACVF and ACF of auto-regressive process, Yule-Walker equations for auto-regressive process, Mixed ARMA models, moving average and auto-regressive representations of mixed ARMA models, Models for Non-stationary Time series, Box-Jenkins Integrated ARIMA models, Stationarity through differencing, other transformations. General linear processes and continuous processes.

Recommended Books:

- Chatfield, C. (2003). The analysis of time series: An introduction (6th ed.). Chapman & Hall: London.
- Wei, W. (1990). Time series analysis: Univariate and multivariate methods. Addison-Wesley publishing company, Inc.
- Box, G.E.P., Jenkins, G.M. & Reinsel, G.C. (2004). Time series analysis: Forecasting and control (3rd ed.). Holden-day: San Francisco.
- Brockwell, P.J., & Davis, R.A. (2002). Introduction to time series and forecasting. (2nd ed.). Springer: New York.



BS (4 Years) for Affiliated Colleges

Code	Subject Title	Cr. Hrs	Semester
STAT-406	Research Methodology	2	VII
Year	Discipline		
4	Statistics		

Course Outline

1. Introduction: Definition of Research, Types and Methods of Conducting Research, Census and Survey, Sampling frame, Types of errors in surveys (coverage, non-response, measurement, errors etc.) and methods of control of such errors, Steps for successful surveys.
2. Types of Surveys: Qualitative and Quantitative survey, Assessments survey, Marketing survey, Evaluation of a survey.
3. Methods for conducting a Survey: Mail surveys, telephone surveys, face to face surveys, and drop off surveys.
4. Sample size: Various methods of sample selection, sample size and its practical difficulties.
5. Constructing a questionnaire for different types of surveys.
6. Scaling Techniques.
7. The analysis of Data.
8. Style and Format of report writing.
9. Preparing the report.

Recommended Books:

- Salant, P. and Dillaman, D.A. "How to conduct your own survey", John Wiley and Sons, Inc. 1994.
- Goode, W. J. and Hatt, P. K. "Methods in Social Research", McGraw-Hill Book Company, Inc.
- Gupta, S. "Research Methodology and Statistical Techniques", Deep & Deep Publication, New Delhi, 1997.
- Dalemius T. "Elements of Survey Sampling" SAREC, Stockholm, 1985.

BS (4 Years) for Affiliated Colleges



Code	Subject Title	Cr. Hrs	Semester
STAT-412	Operation Research	3	VII
Year	Discipline		
4	Statistics		

Course Outlines

Definition and nature of Operations Research (OR). Phases of an OR study. Modeling, constraints, objective and criterion. Problem formulation. Decision Variables, Objective function, Constraints Model building approach to problem solving. Types of models available for OR. Deterministic models, Stochastic Models. Examples of Models. Example of OR applications.

Introduction to linear programming, Graphical solution technique. Simplex Method, Application of Simplex Network.

Recommended Books

- Taha, H.A. "Operations Research, An Introduction". Macmillan Publishing Company, New York, Fifth Edition, 1994.